Numerical solution of ODE & PDE

First order ordinary differential equation (ODE), with initial value

$$\frac{dy}{dx} = f(y(x), x), \quad \text{with initial value } y(x_0) = y_0$$

dy/dx is tangent to solution curve $y = y(x)$ at point x given $x = x_0$, $y = y_0$.

Possible to convert second-order ODE to two coupled first order ODE,

$$\frac{d^2x}{dt^2} = -\omega^2 x \quad \Rightarrow \quad v = \frac{dx}{dt} \text{ and } \frac{dv}{dt} = -\omega^2 x$$

initial conditions being $x(t_0) = x_0$ and $v(t_0) = v_0$.

To solve first order ODEs given an initial condition, the methods we discuss here are

1. Forward (explicit) and Backward (implicit) Euler's method
2. Predictor-Corrector method
3. Runge-Kutta 4th order

Forward (explicit) Euler

Based on Taylor expansion of $f(x_0 + h)$ about x_0 assuming h is small

$$y(x_0 + h) = y(x_0) + h \left.\frac{dy}{dx}\right|_{x_0} + \frac{h^2}{2!} \left.\frac{d^2y}{dx^2}\right|_{x_0} + \cdots \approx y(x_0) + h\, f(y(x_0), x_0) + \mathcal{O}(h^2)$$

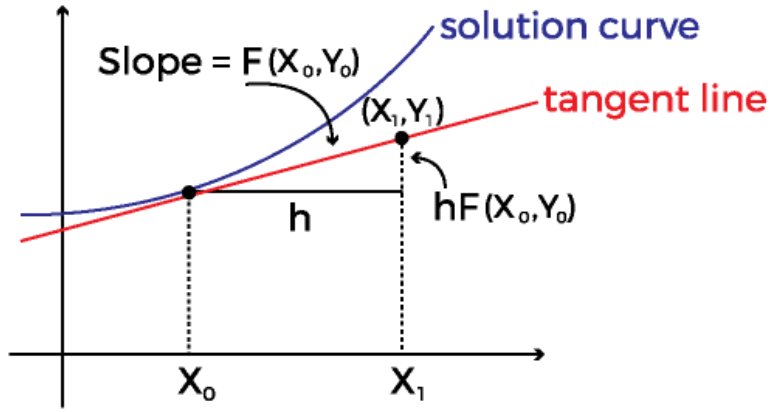
Say $x_1 = x_0 + h$ a small h step away from $x_0 \Rightarrow$ Forward Euler gives

$$y(x_1) = y(x_0) + h\, f\,(y(x_0), x_0) + \mathcal{O}(h^2)$$

Continue from x_1 to $x_2 = x_1 + h$, to x_3 etc. After $n-$th step

$$y(x_n + h) = y(x_n) + h\, f(y(x_n), x_n) \text{ or, equivalently } \boxed{y_{n+1} = y_n + \kappa_1}$$

where $\kappa_1 = hf(y(x_n), x_n)$ implies slope or tangent at the beginning of the interval boundary $\Rightarrow$ dy/dx calculated at earlier point x_n to obtain solution at the end of interval $x_n + h$.



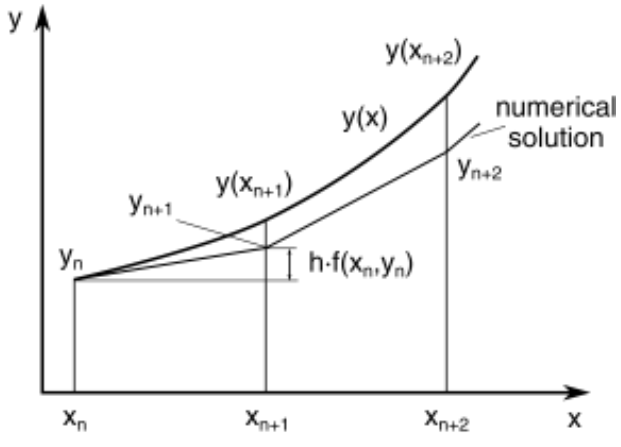
It is Forward because

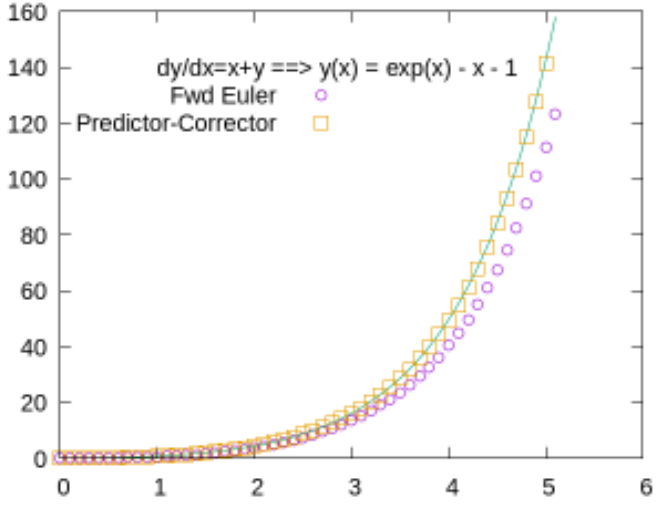
$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} = f(y(x), x)$$

$$\Rightarrow \qquad y(x + \Delta x) \approx y(x) + \Delta x\, f(y(x), x)$$

Forward Euler often returns good approximation to actual solution, but it is extremely slow $\rightarrow$ h has to be small to achieve desired accuracy.

Its biggest problem is stability issue, can easily veer away from solution.





Left plot $\circ$ is for Forward Euler and $\square$ for Predictor-Corrector.

Stability of Forward Euler is more of a problem than its sluggishness.

Backward (implicit) Euler

Re-define dy/dx in terms of *backward derivative*,

$$\begin{aligned}
\frac{\Delta y}{\Delta x} \quad &\approx \quad \frac{y(x) - y(x-\Delta x)}{\Delta x} = f(y(x), x) \quad \Rightarrow \quad y(x) = y(x-\Delta x) + \Delta x\, f(y(x), x) \\
\Rightarrow \quad & \quad y(x+\Delta x) = y(x) + f(y(x+\Delta x), x+\Delta x) \\
\equiv \quad & \quad y(x_n + h) = y(x_n) + h\, f(y(x_n + h), x_n + h)
\end{aligned}$$

$y(x_n + h)$ is determined from the tangent at $x_n + h$, implying strangely that to calculate $y(x_n + h)$ one needs to know $y(x_n + h)$!!

This apparent conflict is resolved by looking it as a linear equation and solve it by Newton-Raphson,

$$y^{\mathrm{NR}}(x + h) = y(x) + h\, f(y^{\mathrm{NR}}(x + h), x + h)$$

Hence, solving the differential equation by

$$y(x_n + h) = y(x_n) + h\, f(y^{\mathrm{NR}}(x_n + h), x_n + h)$$

In spite of having an extra step of Newton-Raphson, the Backward Euler is advantageous because of better stability.

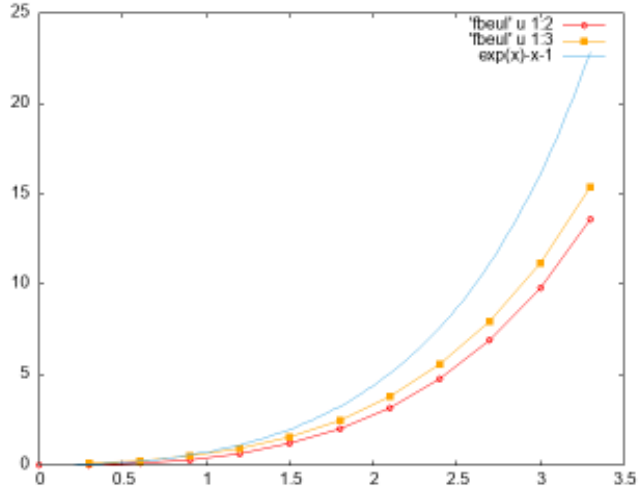
Solve : $\frac{dy}{dx} = x + y$ for $y(x = 0) = 0.0$

Forward Euler : straight forward

$$y_{n+1} = y_n + dx \cdot f(y(x_n), x_n)$$
$$y_{n+1} = y_n + dx \cdot (y_n + x_n)$$

Backward Euler : few lines of calculations

$$y_{n+1} = y_n + dx \cdot f(y(x_{n+1}), x_{n+1})$$
$$y_{n+1} = y_n + dx \cdot (x_{n+1} + y_{n+1})$$
$$y_{n+1}(1 - dx) = y_n + dx \cdot x_{n+1}$$
$$y_{n+1} = \frac{y_n + dx \cdot x_{n+1}}{1 - dx}$$



Predictor-Corrector method

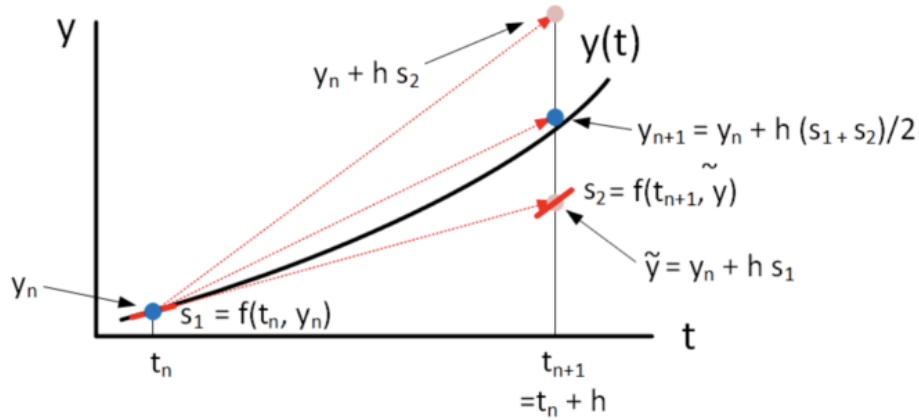
Predicts $y^p(x_n + h)$ using Forward Euler and estimate $f(y^p(x_n + h), x_n + h)$

$$y^p(x_n + h) = y(x_n) + h\, f(y(x_n), x_n) + \mathcal{O}(h^2)$$

Take the average of the two slopes to obtain correct value $y^c(x_n + h)$,

$$y^c(x_n + h) = y(x_n) + \frac{h}{2}\Big[f(y(x_n), x_n) + f(y^p(x_n + h), x_n + h)\Big] \equiv y(x_n) + \frac{1}{2}\Big[\kappa_1 + \kappa_2\Big]$$

where, $\kappa_1 = h\, f(y(x_n), x_n)$, $\kappa_2 = h\, f(y^p(x_n + h), x_n + h)$. The κ_2 denotes predicted slope at the end of interval boundary.



1. Compute slope $\kappa_1 = hf(y(x_n), x_n)$ at x_n.
2. Calculate the predicted $y^p(x_n + h) = y(x_n) + \kappa_1$, hence compute $\kappa_2 = hf(y^p(x_n + h), x_n + h) = h\, f(y(x_n) + \kappa_1, x_n + h)$.
3. Calculate the corrected solution $y^c(x_n + h) = y(x_n) + (\kappa_1 + \kappa_2)/2$

Runge-Kutta method

Runge-Kutta (RK) methods are based on Taylor expansion and give better algorithms for solutions of ODE for same step size and stability.

$$\frac{dy}{dx} = f(y(x),x) \Rightarrow \int_{y_n}^{y_{n+h}} dy = \int_{x_n}^{x_n+h} f(y(x),x)\,dx$$

$$y(x_n+h) = y(x_n) + \int_{x_n}^{x_n+h} f(y(x),x)\,dx$$

Take Midpoint, say, for numerical estimate of the integral

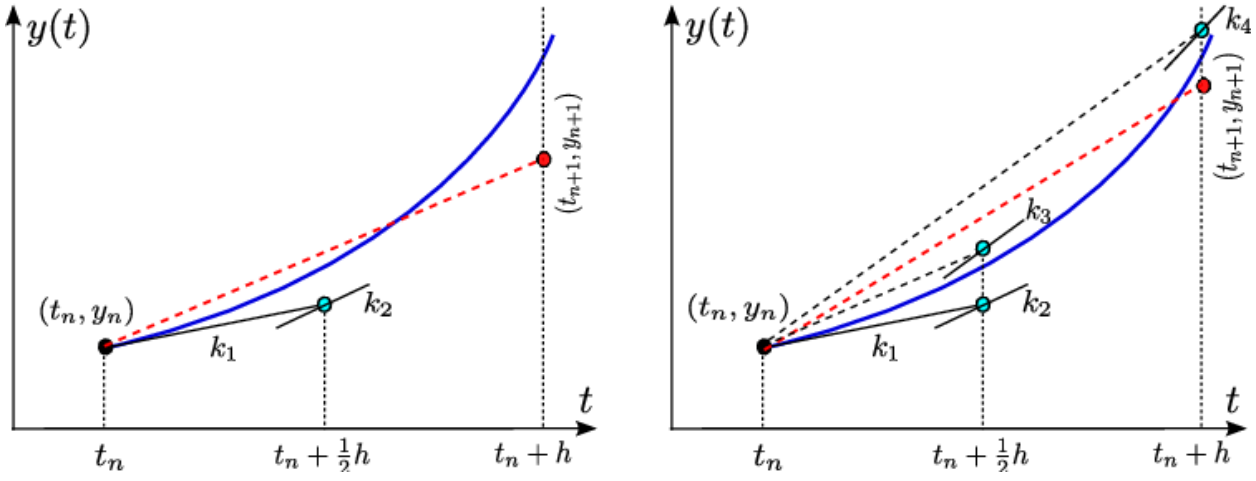
$$\begin{aligned} \overline{x}_n &= \left[(x_n+h)+x_n\right]/2 = x_n + h/2 \\ y(x_n+h) &= y(x_n) + h\,f\left(y(\overline{x}_n),\overline{x}_n\right) + \mathcal{O}(h^3) \\ &= y(x_n) + h\,f\left(y(x_n+h/2),x_n+h/2\right) \end{aligned}$$

Forward Euler: $y(x_n+h/2) = y(x_n) + \frac{h}{2} f(y(x_n),x_n)$

$$\Rightarrow \quad y(x_n+h) = y(x_n) + \overbrace{h\,f\left(y(x_n) + \underbrace{hf(y(x_n),x_n)}_{\kappa_1}/2, x_n+h/2\right)}^{\kappa_2}$$

leading to *second order Runge-Kutta* (RK2)

$$\begin{aligned} \kappa_1 &= h\,f(y(x_n),x_n) \quad \kappa_2 = h\,f(y(x_n)+\kappa_1/2, x_n+h/2) \\ y(x_n+h) &\approx y(x_n) + \kappa_2 + \mathcal{O}(h^3) \end{aligned}$$



Note a couple of points on *RK2* –

1. Difference between one-step methods like Euler's and Predictor-Corrector is addition of an intermediate half-step.
2. Order of error is the maximum error bound of integrator.
3. Trapezoidal rule gives Predictor-Corrector with error $\mathcal{O}(h^3)$.

Next step is using Simpson rule as integrator to develop the famous and by far the most popular method for solving ODE : fourth-order Runge-Kutta (*RK4*).

Unless stated or asked differently, it will always be assumed that RK4 is used to solve ODE.

Consider the integral equation again, this time using Simpson rule,

$$\begin{aligned} y(x_n + h) &= y(x_n) + \int_{x_n}^{x_n+h} f(y(x), x)\, dx \\ &= y(x_n) + \frac{h}{6}\Big[f(y(x_n), x_n) + 4f(y(x_n + h/2), x_n + h/2) + f(y(x_n + h), x_n + h)\Big] \\ &= y(x_n) + \frac{h}{6}\Big[f(y(x_n), x_n) + 2f(y(x_n + h/2), x_n + h/2) + \\ &\qquad 2f(y(x_n + h/2), x_n + h/2) + f(y(x_n + h), x_n + h)\Big] \end{aligned}$$

Essentially, expression for slopes are splitted up at interval midpoint $f(y(x_n + h/2), x_n + h/2)$ into two – one *predicts* the tangent at the interval and the later *corrects* it. Define the following,

$$\begin{aligned} \kappa_1 &= h\, f(y(x_n), x_n) & \kappa_2 &= h\, f(y(x_n) + \kappa_1/2, x_n + h/2) \\ \kappa_3 &= h\, f(y(x_n) + \kappa_2/2, x_n + h/2) & \kappa_4 &= h\, f(y(x_n) + \kappa_3, x_n + h) \end{aligned}$$

Combine these to form the *RK4* solution

$$y(x_n + h) = y(x_n) + \frac{1}{6}\Big(\kappa_1 + 2\kappa_2 + 2\kappa_3 + \kappa_4\Big) + \mathcal{O}(h^5)$$

The error $\mathcal{O}(h^5)$ follows from maximum error bound of Simpson and allow us to use relative coarser interval to arrive at very precise solution.

Coupled ODE

Euler and Runge-Kutta are all applied to first order ODE.

higher order ODE : convert to coupled first order ODE. Take SHO

$$\begin{aligned} \frac{d^2x}{dt^2} &= -\mu\frac{dx}{dt} - \omega^2 x, \quad \text{with} \quad x(t=0) = x_0, \quad v(t=0) = \left.\frac{dx}{dt}\right|_{t=0} = v_0 \\ \Rightarrow \quad v &= \frac{dx}{dt} \quad \text{with} \quad v(t=0) = v_0 \\ \frac{dv}{dt} &= -\mu v - \omega^2 x \quad \text{with} \quad x(t=0) = x_0 \end{aligned}$$

In addition, there can be just a set of coupled first order ODEs. For instance, Lorentz equations and Rössler system,

$$\frac{dx}{dt} = \sigma(y-x), \qquad \frac{dy}{dt} = x(\rho - z) - y, \qquad \frac{dz}{dt} = xy - \beta z$$

$$\frac{dx}{dt} = -y - z, \qquad \frac{dy}{dt} = x + a\,y, \qquad \frac{dz}{dt} = b + z\,(x-c)$$

where chaotic behaviour is observed for $\sigma = 10$, $\rho = 28$, $\beta = 8/3$ for Lorentz and $a = 0.15$, $b = 0.2$, $c = 10$ or $a = 0.2$, $b = 0.2$, $c = 5.7$ for Rössler

The *RK4* for damped SHO takes the following appearance

```
k1x    =   dt*dxdt(x,v,t);
k1v    =   dt*dvdt(x,v,t);

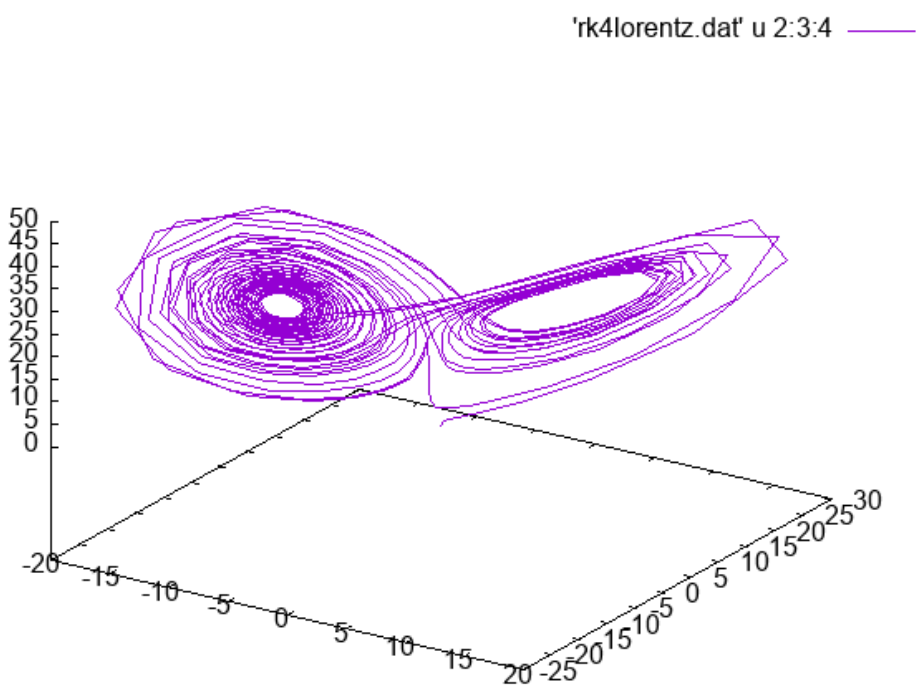
k2x    =   dt*dxdt(x+k1x/2,v+k1v/2,t+dt/2);
k2v    =   dt*dvdt(x+k1x/2,v+k1v/2,t+dt/2);
⋮      =   ⋮
x     +=   (k1x + 2*k2x + 2*k3x + k4x)/6;
v     +=   (k1v + 2*k2v + 2*k3v + k4v)/6;
t     +=   dt;
```

For Lorentz equations, the expressions are similar to SHO

```
k1x    =   dt*dxdt(x,y,z,t);
k1y    =   dt*dydt(x,y,z,t);
k1z    =   dt*dzdt(x,y,z,t);

k2x    =   dt*dxdt(x+k1x/2,y+k1y/2,z+k1z/2,t+dt/2);
k2y    =   dt*dydt(x+k1x/2,y+k1y/2,z+k1z/2,t+dt/2);
k2z    =   dt*dzdt(x+k1x/2,y+k1y/2,z+k1z/2,t+dt/2);
etc.
```

Just for fun : for $\sigma = 10, \rho = 28, \beta = 8/3$, the 3-dim plot of the solution shows the famous Lorentz attractor



Generalizing RK4 to n coupled first order ODE

$$\left.\begin{array}{rcl} \frac{dy_1}{dx} & = & f_1(y_1, y_2, \ldots, y_n, x) \\ \frac{dy_2}{dx} & = & f_2(y_1, y_2, \ldots, y_n, x) \\ & \vdots & \\ \frac{dy_n}{dx} & = & f_n(y_1, y_2, \ldots, y_n, x) \end{array}\right\} \equiv \frac{d\vec{y}}{dx} = \vec{f}(\vec{y}, x)$$

where $\vec{y} = (y_1, y_2, \ldots, y_n)$ and $\vec{f} = (f_1, f_2, \ldots, f_n)$. Vector sign simply implies collection of variables. In such case the RK4 equations take the forms,

$$\begin{aligned}
\vec{\kappa}_1 &= h\,\vec{f}(\vec{y}_i, x_i) \\
\vec{\kappa}_2 &= h\,\vec{f}(\vec{y}_i + \vec{\kappa}_1/2, x_i + h/2) \\
\vec{\kappa}_3 &= h\,\vec{f}(\vec{y}_i + \vec{\kappa}_2/2, x_i + h/2) \\
\vec{\kappa}_4 &= h\,\vec{f}(\vec{y}_i + \vec{\kappa}_3, x_i + h) \\
\vec{y}_{i+h} &= \vec{y}_i + \frac{1}{6}\left[\vec{\kappa}_1 + 2\vec{\kappa}_2 + 2\vec{\kappa}_3 + \vec{\kappa}_4\right]
\end{aligned}$$

where $\vec{y}_i$ are the values at the i-th interval boundary. Above set of equations have to be read only in terms of components and not as vector equations.

Use RK4 to solve the following :

$$\begin{aligned}
\dot{x} &= -5x + 5y \\
\dot{y} &= 14x - 2y - zx \\
\dot{z} &= -3z + xy
\end{aligned}
\qquad\qquad
\begin{aligned}
\dot{x} &= -2y \\
\dot{y} &= x + z^2 \\
\dot{z} &= 1 + y - 2z
\end{aligned}$$

Boundary Value Problem : Shooting method

Many problems in physics are boundary value problems. Like Laplace equation in electrostatics or, more famously, Schrödinger equations.

In boundary value problems, we have conditions specified at two different space (and/or time) points. We can have either

Dirichlet condition : $y(x_0) = Y_0$ and $y(x_N) = Y_n$

Neumann condition : $y'(x_0) = Y'_0$ and $y'(x_N) = Y'_N$

$$\frac{d^2y}{dx^2} = f(x, y, y') \quad \text{where} \quad a \leq x \leq b \quad \text{and} \quad y(a) = \alpha,\ y(b) = \beta$$

$$\Rightarrow \quad \frac{dy}{dx} = z \quad \text{with} \quad y(a) = \alpha$$

$$\frac{dz}{dx} = \frac{d^2y}{dx^2} = f(x, y, z) \quad \text{with} \quad z(a) = \zeta_h \quad (\textit{guess})$$

the slope $z(a) = \zeta_h$ at $x = a$ is a guess. Solve the coupled ODEs with initial values $y(a) = \alpha$ and $z(a) = \zeta_h$.

Solution obtained at end point x_N is compared with the boundary condition $y(b) = \beta$. If $y_{\zeta_h}(b) = \beta$ within tolerance then the ODE is solved.

Suppose $y_{\zeta_h}(b) \neq \beta$ but $y_{\zeta_h}(b) > \beta$. Change the guess initial value $\zeta_h \to \zeta_l$ of course and solve it again.

Unless the choice lands bang on the solution, choose ζ_l such that $y_{\zeta_l}(b) < \beta$ implying

$$\zeta_l < z(a) < \zeta_h$$

Use Lagrange's interpolation formula to choose the next $z(a) = \zeta$

$$\zeta = \zeta_l + \frac{\zeta_h - \zeta_l}{y_{\zeta_h}(b) - y_{\zeta_l}(b)} \Big(y(b) - y_{\zeta_l}(b) \Big)$$

$z(a) = \zeta$ is our new guess value for initial slope and chances are this choice will lead us to the solution of the ODE *i.e.* $y_\zeta(b) \approx \beta$.

If not, go through the above procedure until $y_\zeta(b)$ converges to β.

Let us study the following example,

$$\frac{d^2y}{dx^2} = 2y \quad \text{with} \quad y(x = 0.0) = \alpha = 1.2, \quad y(x = 1.0) = \beta = 0.9$$

Analytical solution of the above ODE is

$$y(x) = c_1\, e^{\sqrt{2}\,x} + C_2\, e^{-\sqrt{2}\,x}, \quad \text{where} \quad C_1 = 0.157, \; C_2 = 1.043$$

Let our initial guess slope is $z(x = 0.0) = -1.5$. RK4 returns

$$y(x = 1.0) = 0.5614 < \beta = 0.9 \quad \Rightarrow \quad \zeta_l = -1.5,\ y_{\zeta_l}(1.0) = 0.5614$$

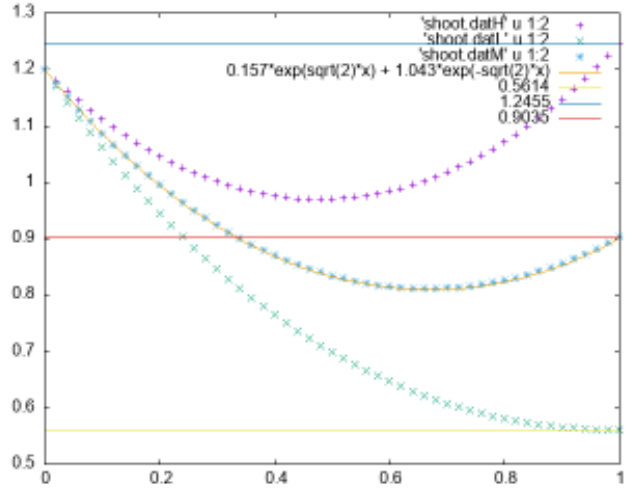
Next try $z(x = 0.0) = -1.0$,

$$y(x = 1.0) = 1.2455 > \beta = 0.9 \quad \Rightarrow \quad \zeta_h = -1.0,\ y_{\zeta_h}(1.0) = 1.2455$$

Use Lagrange's linear interpolating formula,

$$\zeta = -1.5 + \frac{-1.0 - (-1.5)}{1.2455 - 0.5614} \times \Big(0.9 - 0.5614\Big) = -1.2525$$

Using $z(x = 0.0) = -1.2525$, we obtain $y_\zeta(x = 1.0) = 0.9035 \approx \beta = 0.9$ thus solving the ODE. A graphical view of the process



Finite element method

Discretization of derivative using Taylor's series. Let us solve,

$$y''(x) = f(x) \quad \text{with} \quad a \leq x \leq b, \quad \text{where,} \quad y(x = a) = \alpha, \quad y(x = b) = \beta$$

Domain discretization : Divide $[a, b]$ into N equal sub-intervals

$$b - a = Nh \quad \Rightarrow \quad x_n = a + nh \quad \text{where} \quad n = 0, 1, \ldots, N$$

Known $y(a) \equiv y_0 @ x_0 = \alpha$, $y(b) \equiv y_N @ x_N = \beta$ and unknown $y_1, y_2, \ldots, y_{N-1}$

$$y''(x_n) \approx \frac{y_{n+1} - 2y_n + y_{n-1}}{h^2} \quad \Rightarrow \quad y_{n-1} - 2y_n + y_{n+1} = h^2 f(x_n)$$

At boundaries $n = 0,\ N$ and for any interior nodes $x_n, \quad n \neq 0,\ N$

$$\begin{aligned}
y_0 - 2y_1 + y_2 &= h^2 f(x_1) && \Rightarrow \ 2y_1 - y_2 = -h^2 f_1 + \alpha \\
y_{N-2} - 2y_{N-1} + y_N &= h^2 f(x_{N-1}) && \Rightarrow \ -y_{N-2} + 2y_{N-1} = -h^2 f_{N-1} + \beta \\
y_{n-1} - 2y_n + y_{n+1} &= h^2 f(x_n) && \Rightarrow \ -y_1 + 2y_2 - y_3 = -h^2 f_2 \\
& && \Rightarrow \ -y_2 + 2y_3 - y_4 = -h^2 f_3 \\
& && \Rightarrow \ \vdots = \vdots
\end{aligned}$$

Stacking these equations, we arrived at tridiagonal, symmetric, positive definite matrix

$$\begin{pmatrix} 2 & -1 & 0 & 0 & \cdots & & 0 \\ -1 & 2 & -1 & 0 & \cdots & & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & -1 & 2 & -1 \\ 0 & \cdots & \cdots & & & -1 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_{N-2} \\ y_{N-1} \end{pmatrix} = \begin{pmatrix} -h^2 f_1 + \alpha \\ -h^2 f_2 \\ \vdots \\ -h^2 f_{N-2} \\ -h^2 f_{N-1} + \beta \end{pmatrix}$$

Standard method to solve for such a tridiagonal system is Cholesky decomposition subsequently using forward and backward substitution.

Partial differential equations

Many equations in physics are partial differential equations – Maxwell equations, Laplace and Poisson equations, wave equations, Schrödinger equation, diffusion equation and so on.

General linear partial differential equation in 2-variables $x,\, y \,\to\, u(x,y)$ *i.e.* second-order in two independent variables reads,

$$A(x,y)\,\frac{\partial^2 u}{\partial x^2} + B(x,y)\,\frac{\partial^2 u}{\partial x \partial y} + C(x,y)\,\frac{\partial^2 u}{\partial y^2} = F\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)$$

Depending on coefficients $A,\, B,\, C$, the PDE are classified as

1. elliptic if $B^2 - 4AC < 0$
2. parabolic if $B^2 - 4AC = 0$
3. hyperbolic if $B^2 - 4AC > 0$

$B = C = 0$ is the $1+1$-dim diffusion equation $D\,u_{xx} = \dot{u}$ is parabolic, whereas $B = 0$ and $AC < 0$ is $2+1$-dim wave equation is elliptic.

$$\ddot{u} = c^2\left(u_{xx} + u_{yy}\right)$$

What would be the class of Maxwell's equations?

Most common method employed to solve PDE is *finite difference*, converting derivatives to differences, similar to FEM.

For simplicity consider only two dof, x and y, the function is $u(x, y)$

Two categories of method exist – explicit (forward difference) and implicit (backward difference).

$$\text{forward} \quad \left.\frac{\partial u}{\partial x}\right|_y = \frac{u(x+\Delta x, y) - u(x, y)}{\Delta x} + \mathcal{O}(\Delta x) = u_x$$

$$\text{forward} \quad \left.\frac{\partial u}{\partial y}\right|_x = \frac{u(x, y+\Delta y) - u(x, y)}{\Delta y} + \mathcal{O}(\Delta y) = u_y$$

$$\text{backward} \quad \left.\frac{\partial u}{\partial x}\right|_y = \frac{u(x, y) - u(x-\Delta x, y)}{\Delta x} + \mathcal{O}(\Delta x) = u_x$$

$$\text{backward} \quad \left.\frac{\partial u}{\partial y}\right|_x = \frac{u(x, y) - u(x, y-\Delta y)}{\Delta y} + \mathcal{O}(\Delta y) = u_y$$

$$\text{symmetric} \quad \left.\frac{\partial u}{\partial x}\right|_y = \frac{u(x+\Delta x, y) - u(x-\Delta x, y)}{2\Delta x} + \mathcal{O}(\Delta x^2)$$

$$\text{symmetric} \quad \left.\frac{\partial u}{\partial y}\right|_x = \frac{u(x, y+\Delta y) - u(x, y-\Delta y)}{2\Delta y} + \mathcal{O}(\Delta y^2)$$

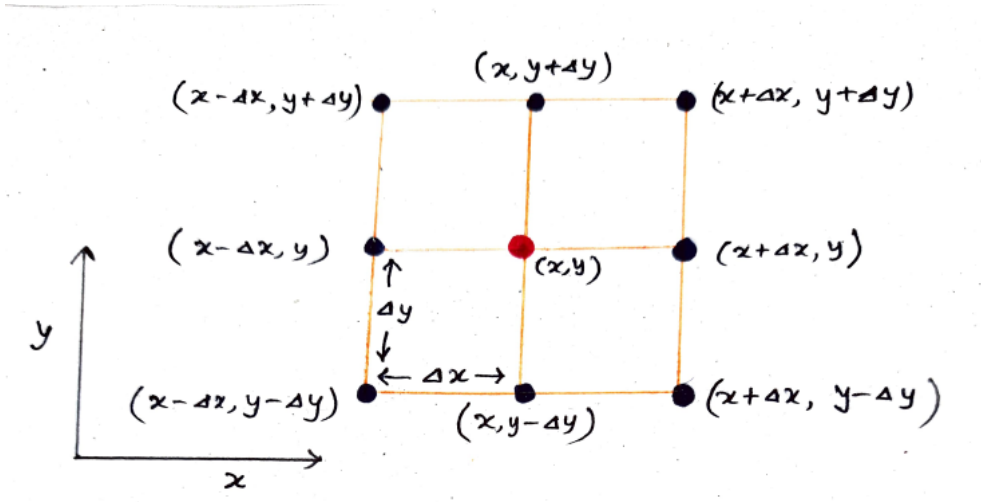
$$u_{xx} = \left.\frac{\partial^2 u}{\partial x^2}\right|_y = \frac{u(x+\Delta x, y) + u(x-\Delta x, y) - 2u(x, y)}{\Delta x^2} + \mathcal{O}(\Delta x^2)$$

and similarly for u_{yy}

Second order partial derivative, u_{xy}, upto $\mathcal{O}(\Delta x^2 + \Delta y^2$

$$u_{xy} = \frac{u(x+\Delta x, y+\Delta y) + u(x-\Delta x, y-\Delta y) - u(x+\Delta x, y-\Delta y) - u(x-\Delta x, y+\Delta y}{4\Delta x \Delta y}$$

Both first and second order partial difference require the point itself and at most two nearest neighbor points. It forms a 3-point stencil.



Most often used is 5-point stencil that requires the point itself and four nearest neighbors. For instance, u_x is, without showing the y coordinate

$$u_x = \frac{-u(x+2\Delta x) + 8u(x+\Delta x) - 8u(x-\Delta x) + u(x-2\Delta x)}{12\Delta x} + \mathcal{O}(\Delta x^4)$$

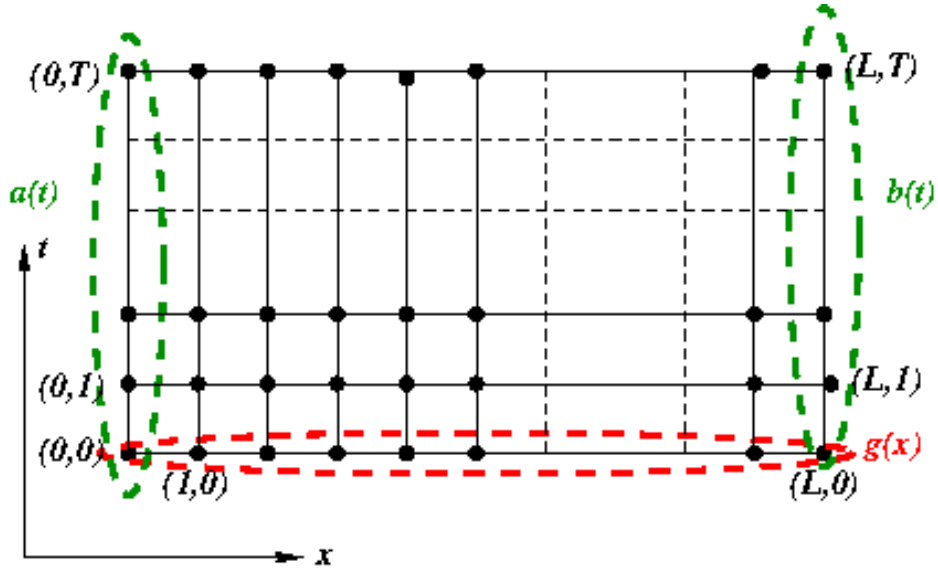
$$u_{xx} = \frac{-u(x+2\Delta x) + 16u(x+\Delta x) - 30u(x) + 16u(x-\Delta x) - u(x-2\Delta x)}{12\Delta x^2} + \mathcal{O}(\Delta x^5)$$

Consider explicit / forward scheme in 1+1 dim diffusion / heat equation,

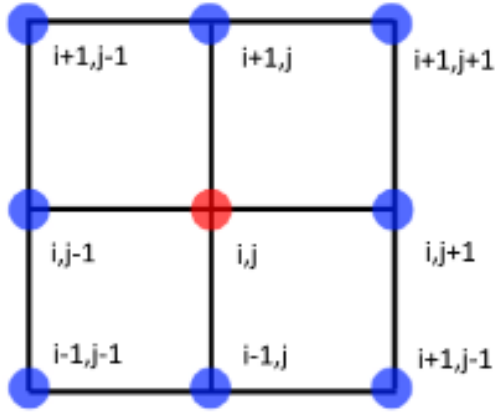
$$\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial u(x,t)}{\partial t} \quad \text{i.e.} \quad u_{xx} = u_t$$

boundary conditions at $t = 0$ and at later time $t > 0$,

$$\begin{aligned} u(x,0) &= g(x) \quad \text{for} \quad 0 < x < L \\ u(0,t) &= a(t) \quad \text{for} \quad t \geq 0 \\ u(L,t) &= b(t) \quad \text{for} \quad t \geq 0 \end{aligned}$$



Discretize space and time such that $\Delta x = h_x$, $\Delta t = h_t$ are differences between two space and time points respectively.



Position after i steps and time at j step are given by

$$\begin{cases} x_i = ih_x & 0 \le i \le n+1 \\ t_j = jh_t & j \ge 0 \end{cases}$$

Discretized forward derivatives for explicit scheme are

$$\begin{aligned} u_t &\approx \frac{u(x, t+h_t) - u(x,t)}{h_t} \equiv \frac{u(x_i, t_j + h_t) - u(x_i, t_j)}{h_t} = \frac{u_{i,j+1} - u_{i,j}}{h_t} \\ u_{xx} &\approx \frac{u(x+h_x, t) + u(x-h_x, t) - 2u(x,t)}{h_x^2} \\ &\equiv \frac{u(x_i + h_x, t_j) + u(x_i - h_x, t_j) - 2u(x_i, t_j)}{h_x^2} = \frac{u_{i+1,j} + u_{i-1,j} - 2u_{i,j}}{h_x^2} \end{aligned}$$

Define $\alpha = h_t / h_x^2$.

The diffusion / heat equation results in explicit scheme

$$\begin{aligned} \frac{u_{i,j+1} - u_{i,j}}{h_t} &= \frac{u_{i+1,j} + u_{i-1,j} - 2u_{i,j}}{h_x^2} \\ u_{i,j+1} &= = \alpha \left(u_{i+1,j} + u_{i-1,j}\right) + (1 - 2\alpha)\, u_{i,j} \end{aligned}$$

Given initial values $u_{i,0} = g(x_i)$, after one time step we get $u_{i,1}$,

$$\begin{aligned} u_{i,1} &= \alpha \left(u_{i+1,0} + u_{i-1,0}\right) + (1 - 2\alpha)\, u_{i,0} \\ &= \alpha \left[g(x_{i+1}, 0) + g(x_{i-1}, 0)\right] + (1 - 2\alpha)\, g(x_i, 0) \end{aligned}$$

For simplicity and without loss of generality, consider $a(t) = b(t) = 0$ implying $u_{0,j} = u_{L=n+1,j} = 0$ where the interval $[0, L]$ is divided in n parts.

Then a vector V_j at the time $t_j = j\, h_t$ is defined as,

$$V_j = \begin{pmatrix} u_{1,j} \\ u_{2,j} \\ \vdots \\ u_{n,j} \end{pmatrix}$$

with $V_0 \equiv u_{i,0} = g(x_i)$.

Therefore, the solution at a given time slice V_{j+1} proceeds as

$$V_{j+1} = \mathbf{A}\, V_j \quad \text{where} \quad \mathbf{A} = \begin{pmatrix} 1-2\alpha & \alpha & 0 & 0 & \cdots \\ \alpha & 1-2\alpha & \alpha & 0 & \cdots \\ & \vdots & \ddots & \vdots & \vdots \\ 0 & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & \alpha & 1-2\alpha \end{pmatrix}$$

This implies that the solution is evolving in time using matrix-vector multiplication till the end of time, say $u_{i,T}$

$$V_{j+1} = \mathbf{A} V_j = \mathbf{A}^2 V_{j-1} = \cdots = \mathbf{A}^{j+1} V_0$$

Like explicit Euler, it too has weak stability condition given by $\alpha = h_t/h_x^2 \leq 0.5$ and is overcome by implicit scheme using backward derivative in time, keeping second derivative unchanged.

$$\begin{aligned} u_t &\approx \frac{u_{i,j} - u_{i,j-1}}{h_t} \\ u_{i,j-1} &= -\alpha \left(u_{i+1,j} + u_{i-1,j}\right) + (1+2\alpha)\, u_{i,j} \end{aligned}$$

The corresponding evolution equation for implicit scheme is

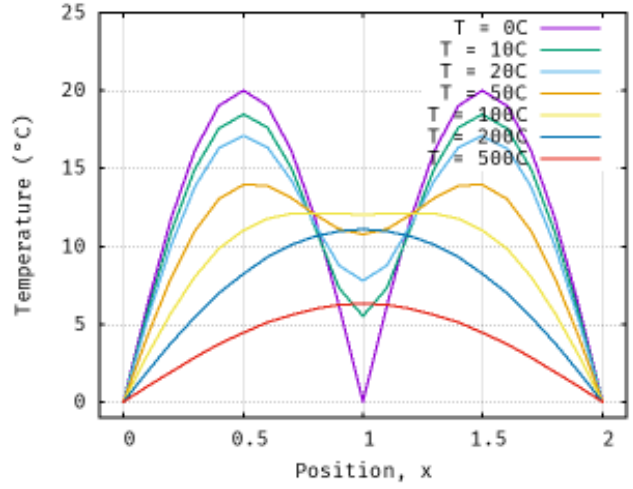
$$V_{j-1} \quad = \quad \mathbf{A}\, V_j \quad \text{where} \quad \mathbf{A} = \begin{pmatrix} 1+2\alpha & -\alpha & 0 & 0 & \cdots \\ -\alpha & 1+2\alpha & -\alpha & 0 & \cdots \\ & \vdots & \ddots & \vdots & \vdots \\ 0 & \vdots & & & \\ 0 & 0 & \cdots & -\alpha & 1+2\alpha \end{pmatrix}$$

$$V_j \quad = \quad \mathbf{A}^{-1} V_{j-1} = \mathbf{A}^{-2} V_{j-2} = \cdots = \mathbf{A}^{-j} V_0$$

Problem : Solve the 1-dimension heat equation $u_{xx} = u_t$ over a metal rod of length 2 units, with the initial conditions,

$$u(0,t) = 0^o\mathrm{C} = u(2,t) \qquad \text{for} \quad 0 \le t \le 4$$
$$u(x,0) = 20\,|\sin(\pi x)|^o\mathrm{C} \qquad \text{for} \quad 0 \le x \le 2$$

Use *explicit scheme* taking number of position grid $nx = 20$ and time grid $nt = 5000$. Show the temperature profile across the length of the rod at time steps 0, 10, 20, 50, 100, 200, 500 and 1000 in a plot. It should look something like



Symmetric derivative : Crank-Nicolson

u_{xx} is approximated by the average of u_{xx} at two different time slices $j-1$ and j (or, alternatively j and $j+1$)

$$u_{xx} = \frac{1}{2\Delta x^2}\Big[\,(u_{i+1,j-1} + u_{i-1,j-1} - 2u_{i,j-1}) + (u_{i+1,j} + u_{i-1,j} - 2u_{i,j})\,\Big]$$

Using this in heat equation $u_{xx} = u_t$ and backward discretised u_t

$$\begin{aligned}
\frac{u_{i,j} - u_{i,j-1}}{\Delta t} &= \frac{1}{2\Delta x^2}\Big[\,(u_{i+1,j-1} + u_{i-1,j-1} - 2u_{i,j-1}) + \\
&\qquad (u_{i+1,j} + u_{i-1,j} - 2u_{i,j})\,\Big] \\
2(u_{i,j} - u_{i,j-1}) &= \alpha\Big[(u_{i+1,j-1} + \ldots - 2u_{i,j-1}) + (u_{i+1,j} + \ldots - 2u_{i,j})\Big]
\end{aligned}$$

Re-arranging terms with subscripts j and $j-1$ on either side

$$\begin{aligned}
& 2(1+\alpha)u_{i,j} + \alpha(-u_{i+1,j} - u_{i-1,j}) = \alpha(u_{i+1,j-1} + u_{i-1,j-1}) + 2(1-\alpha)u_{i,j-1} \\
\Rightarrow \quad & 2u_{i,j} + \alpha\underbrace{(-u_{i-1,j} + 2u_{i,j} - u_{i+1,j})}_{\mathbb{B}} = 2u_{i,j-1} + \alpha\underbrace{(u_{i-1,j-1} - 2u_{i,j-1} + u_{i+1,j-1})}_{\mathbb{B}}
\end{aligned}$$

Strip the j indexed component out from $u_{n,j}$ and write that as V_j and similarly for $j-1$. Then, the above equation can be written in matrix notation as,

$$\begin{aligned}
\left(2\mathbb{I} + \alpha \mathbf{B}\right) V_j &= \left(2\mathbb{I} - \alpha \mathbf{B}\right) V_{j-1} \\
\text{or,} \quad V_j &= \left(2\mathbb{I} + \alpha \mathbf{B}\right)^{-1} \left(2\mathbb{I} - \alpha \mathbf{B}\right) V_{j-1} \\
\text{where,} \quad \mathbf{B} &= \begin{pmatrix} 2 & -1 & 0 & 0 & \cdots \\ -1 & 2 & -1 & 0 & \cdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}
\end{aligned}$$

Crank-Nicolson method is, therefore, a combination of explicit and implicit schemes. The method is known to converge for all values of α and, hence, very useful where coarse graining in any of the space or time direction is needed.